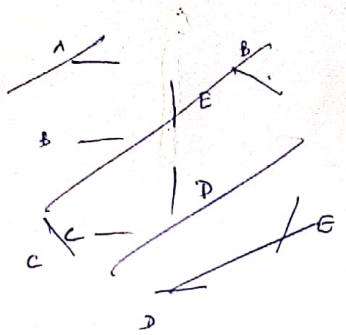
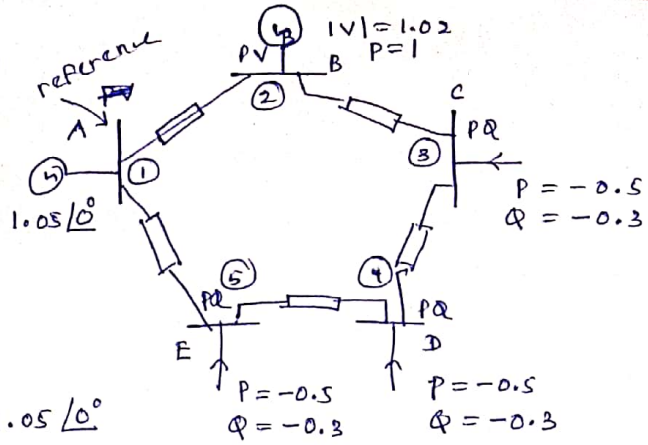
EE 354 – 2017

Q1) HT



1.1)



$Y_{AB} = 1-4j$

$Y_{AE} = 2-8j$

$Y_{ED} = 1-4j$

$Y_{DC} = 1-4j$

$Y_{CB} = 2-8j$

① ⇒ $V_1 = 1.05\angle 0^\circ$

② ⇒ $|V_2| = 1.02$, $P_2 = 1$

③ ⇒ $P_3 = -0.5$, $Q_3 = -0.3$

④ ⇒ $P_4 = -0.5$, $Q_4 = -0.3$

⑤ ⇒ $P_5 = -0.5$, $Q_5 = -0.3$

PV → 1

PQ → 3

1.2)

$$Y = \begin{array}{c} \\ A \\ B \\ C \\ D \\ E \end{array} \begin{array}{c} \begin{array}{ccccc} A & B & C & D & E \end{array} \\ \begin{bmatrix} 12.37\angle -75.96^\circ & 4.12\angle 104.04^\circ & 0 & 0 & 8.24\angle 104.04^\circ \\ 3-12j & -1+4j & & & -2+8j \\ 4.12\angle 104.04^\circ & 12.37\angle -75.96^\circ & & & \\ -1+4j & 3-12j & -2+8j & 0 & 0 \\ 0 & 8.24\angle 104.04^\circ & 12.37\angle -75.96^\circ & & \\ & -2+8j & 3-12j & -1+4j & 0 \\ 0 & 0 & 4.12\angle 104.04^\circ & 8.24\angle -104.04 & \\ & & -1+4j & 2-8j & -1+4j \\ -2+8j & 0 & 0 & -1+4j & 12.37\angle -75.96^\circ \\ & & & & 3-12j \end{bmatrix} \end{array}$$

1.3)

$$V_K^i = \frac{1}{Y_{KK}}\left[\frac{P_K - jQ_K}{V_K^{*\,i-1}} - \sum_{S=1}^{K-1} Y_{KS} V_S^i - \sum_{S=K+1}^{N} Y_{KS} V_S^{i-1}\right]$$

1.4) $i = 7 \Rightarrow V_A^7 = 1.05\angle 0^\circ = V_1^7$ ← reference bus

$V_B^7 = 1.02\angle 5^\circ = V_2^7$ $\qquad P_2 = P_B = 1$

$V_C^7 = 0.99\angle 1^\circ = V_3^7$ $\qquad Q_2 = Q_B = ?$

$V_D^7 = 0.98\angle 1^\circ = V_4^7$

$V_E^7 = 0.99\angle 2^\circ = V_5^7$

$i = 8$, $K = 2$, $N = 5$

$$V_B^8 = V_2^8 = \frac{1}{Y_{22}}\left[\frac{P_2 - jQ_2}{V_2^{*\,7}} - \sum_{S=1}^{1} Y_{2S} V_S^8 - \sum_{S=3}^{5} Y_{2S} V_S^7\right]$$

$$V_B^8 = V_2^8 = \frac{1}{Y_{22}}\left[\frac{P_2 - jQ_2}{V_2^{*\,7}} - Y_{21} V_1^8 - Y_{23} V_3^7 - \underbrace{Y_{24}}_{0} V_4^7 - \underbrace{Y_{25}}_{0} V_5^7\right]$$

$N = 5$

$i = 8$

$K = 1$; $V_1^8 = V_A^8 = \frac{1}{Y_{11}}\left[\frac{P_1 - jQ_1}{V_1^{*\,7}} - \sum_{S=2}^{5} Y_{1S} V_S^7\right]$

$$V_1^8 = \frac{1}{Y_{11}}\left[\frac{P_1 - jQ_1}{V_1^{*\,7}} - Y_{12} V_2^7 - Y_{23} V_3^7 - Y_{24} V_4^7 - Y_{25} Y_5^7\right]$$

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$$Q_K = \sum_{s=1}^{N} |V_K Y_{Ks} V_s| \sin(\delta_K - \delta_s - \theta_{Ks})$$

$K=2, \quad N=5;$

$$Q_2 = \sum_{s=1}^{5} |V_2 Y_{2s} V_s| \sin(\delta_2 - \delta_s - \theta_{2s})$$

$$= |V_2 Y_{21} V_1| \sin(\delta_2 - \delta_1 - \theta_{21}) + |V_2 Y_{22} V_2| \sin(-\theta_{22}) + |V_2 Y_{23} V_3| \sin(\delta_2 - \delta_3 - \theta_{23})$$

$$+ |V_2 \underbrace{Y_{24}}_{0} V_4| \sin(\delta_2 - \delta_4 - \theta_{24}) + |V_2 \underbrace{Y_{25}}_{0} V_5| \sin(\delta_2 - \delta_5 - \theta_{25})$$

$$= |1.02 \times 4.12 \times 1.05| \sin(5 - 0 - 104.04^\circ) + |1.02 \times 12.37 \times 1.02| \sin(5 - 5 + 75.96^\circ)$$

$$+ |1.02 \times 8.24 \times 0.99| \sin(5 - 1 - 104.04^\circ)$$

$$= -4.3577 + 12.4853 - 8.1933$$

$$= -0.0657$$

$$V_B^8 = V_2^8 = \frac{1}{12.37 \angle -75.96^\circ} \left[\frac{1 + j\,0.0657}{1.02 \angle -5^\circ} - (-1+4j)(1.05 \angle 0^\circ) - (-2+8j)(0.99 \angle 1^\circ) \right]$$

$$V_B^8 = 1.021 \angle 5.087^\circ \text{ pu} \quad //$$

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(1.5) $J = \begin{array}{c|cccc} & 2 & 3 & 4 & 5 \\ \hline 2 & \partial P_2/\partial \theta & & & \\ 3 & \partial P_3/\partial \theta & & & \\ 4 & & & & \\ 5 & & & & \end{array}$

(1.5)
$$\begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \Delta P_4 \\ \Delta P_5 \\ \Delta Q_3 \\ \Delta Q_4 \\ \Delta Q_5 \end{bmatrix} = \begin{bmatrix} \frac{\partial P_2}{\partial \delta_2} & \frac{\partial P_2}{\partial \delta_3} & \frac{\partial P_2}{\partial \delta_4} & \frac{\partial P_2}{\partial \delta_5} & \frac{\partial P_2}{\partial V_3} & \frac{\partial P_2}{\partial V_4} & \frac{\partial P_2}{\partial V_5} \\ \frac{\partial P_3}{\partial \delta_2} & \frac{\partial P_3}{\partial \delta_3} & \frac{\partial P_3}{\partial \delta_4} & \frac{\partial P_3}{\partial \delta_5} & \frac{\partial P_3}{\partial V_3} & \frac{\partial P_3}{\partial V_4} & \frac{\partial P_3}{\partial V_5} \\ & & & & & & \\ & & \ddots & & & & \\ & & & & & & \end{bmatrix} \begin{bmatrix} \Delta \delta_2 \\ \Delta \delta_3 \\ \Delta \delta_4 \\ \Delta \delta_5 \\ \Delta V_3 \\ \Delta V_4 \\ \Delta V_5 \end{bmatrix}$$

(1.6) after 1st Iteration;

$\Delta V_C = \Delta V_3 = 0.1$ $\quad$ $\Delta \delta_B = \Delta \delta_2 = 1^\circ$

$\Delta V_D = \Delta V_4 = -0.1$ $\quad$ $\Delta \delta_C = \Delta \delta_3 = 3^\circ$

$\Delta V_E = \Delta V_5 = -0.1$ $\quad$ $\Delta \delta_D = \Delta \delta_4 = -4^\circ$

$\Delta \delta_E = \Delta \delta_5 = -2^\circ$

$\Delta V = V_{actual} - V_{\cancel{calculated}\ initial}$

$V_{actual} = \Delta V + V_{initial}$

$V_A' = 1.05 \angle 0^\circ$ pu

$V_B' = 1.02 \angle 1^\circ$ pu

$V_C' = 1.1 \angle 3^\circ$ pu

$V_D' = 0.9 \angle -4^\circ$ pu

$V_E' = 0.9 \angle -2^\circ$ pu

& $\delta_B = 1^\circ + 0^\circ = 1^\circ$

$\delta_C = 3^\circ + 0^\circ = 3^\circ$

$\delta_D = -4^\circ + 0^\circ = -4^\circ$

$\delta_E = -2^\circ + 0^\circ = -2^\circ$

$V_C = 0.1 + 1 = 1.1$

$V_D = -0.1 + 1 = 0.9$

$V_E = -0.1 + 1 = 0.9$

(Q2) 2.3) $V_{LL} = 400$ V $\quad$ Y connected

$P = 4$ $\quad$ $I_{A,full\ load} = 100$ A @ 0.85 PF lagging

$f_e = 50$ Hz $\quad$ $P_{F\&W} = 5$ kW

$P_{core} = 5$ kW

$X_S = 1.5\ \Omega$

$R_A = 0.1\ \Omega$

a) $P_{out} = \sqrt{3}\, V_L I_L \cos\theta = \sqrt{3} \times 400 \times 100 \times 0.85 = 58.889$ kW

b) $P_{in} = P_{out} + P_{F\&W} + P_{core} + P_{cu}$

$= 58.889 + 5 + 5 + \frac{3 \times 100^2 \times 0.1}{1000}$

$= 71.889$ kW

c) $\eta = \frac{58.889}{71.889} \times 100\% = 81.92\%$

d) $P_{in} = \tau_{app}\, \omega_{sync}$ $\quad$ $n_{sync} = \frac{120 \times 50}{4} = 1500$ rpm

$\tau_{app} = \frac{71.889 \times 10^3}{50\pi}$ $\quad$ $\omega_{sync} = \frac{1500 \times 2\pi}{60} = 50\pi$ rad s^{-1}

$\tau_{app} = 457.66$ Nm

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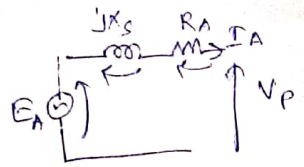
~~$P_{conv} = P_{in}$~~

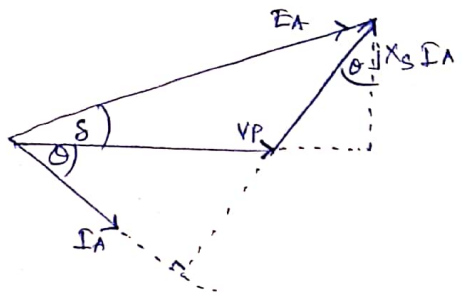
$P_{conv} = \tau_{ind} \times \omega_{sync}$

$$\tau_{ind} = \frac{P_{out} + P_{cu}}{\omega_{sync}} = \frac{\left(58.889 + \frac{3 \times 100^2 \times 0.1}{1000}\right) \times 10^3}{50\pi} = 393.998 \text{ Nm}$$

(e) I_A will be zero

$\therefore$ Terminal voltage $= E_A$





$E_A = V_P + I_A(jX_s + \cancel{R_A})$

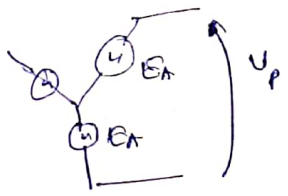
0

$$E_A^2 = (X_s I_A \cos\theta)^2 + (V_P + X_s I_A \sin\theta)^2 \quad \left[\text{by neglecting } R_A\right]$$

$$= (1.5 \times 100 \times 0.85)^2 + \left(\frac{480}{\sqrt{3}} + 1.5 \times 100 \times \sin\theta\right)^2$$

$$= 16256.25 + 126839.64$$

$$E_A = 378.28 \text{ V}$$



$\therefore$ Terminal voltage $= V_P = \sqrt{3}\, E_A$

$= 655.2$ V

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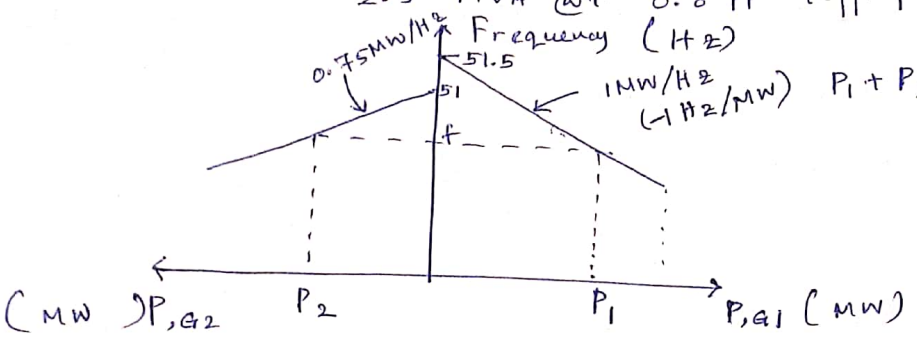
Question 3

3.2) G1 ⇒ $f_{nl} = 51.5$ Hz , slope = 1 MW/Hz

G2 → $F_{nl} = 51.0$ Hz, slope = 0.75 MW/Hz

(a) Load = 2.5 MVA @ 0.8 PF lagging



$P_1 + P_2 = 2.5 \times 0.8 = 2$ MW — (3)

G1 ⇒ $\frac{51.5 - f}{0 - P_1} = \frac{-1 \text{ Hz}}{1 \text{ MW}} = -1$

$+P_1 = 51.5 - f$ (1)

~~$P_1 = f - 51.5$ — (1)~~

G2 ⇒ $\frac{51 - f}{0 - P_2} = \frac{-1 \text{ Hz}}{0.75 \text{ MW}}$

$+P_2 = 0.75(51 - f)$ — (2)

~~$P_2 = 0.75(f - 51)$ — (2)~~

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①, ②, ③ $\Rightarrow$ $(51.5 - f) + 0.75(51 - f) = P_{TOT}$

$89.75 - 1.75f = P_{TOT}$

$$f = \frac{89.75 - P_{TOT}}{1.75}$$

$$f = \frac{89.75 - 2}{1.75} = 50.14 \text{ Hz}$$

$P_1 = 51.5 - 50.14 = 1.36$ MW //

$P_2 = 0.75(51 - 50.14) = 0.64$ MW

(b) 8-9 $\Rightarrow$ $f = 89.75 - (2 - 0.5) / 1.75 = 50.43$ Hz //

$P_1 = 51.5 - 50.43 = 1.07$ MW //

$P_2 = 0.75(51 - 50.43) = 0.43$ MW //

9-10 $\Rightarrow$ $f = 89.75 - (2 - 1) / 1.75 = 50.71$ Hz

$P_1 = 51.5 - 50.71 = 0.79$ MW //

$P_2 = 0.75(51 - 50.71) = 0.21$ MW //

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$17 - 17 \Rightarrow f = 89.75 - (2 - 1.5)/1.75 = 51\ Hz$

$P_1 = 51.5 - 51 = 0.5\ MW$

$P_2 = 0.75\ (51 - 51) = 0\ MW \leftarrow$ no contribution.

$17 - 12 \Rightarrow f = 89.75 - (2 - 1.6)/1.75 = 51.06\ Hz$

$P_1 = 51.5 - 51.06 = 0.44\ MW$

$P_2 = 0.75(51 - 51.06) < 0 \rightarrow$ cannot supply power.

c) $P_{load,\ new} = 2 \times \frac{150}{100} = 3\ MW$

$f = 89.75 - (3 - 1.5)/1.75 = 50.43\ Hz$

To restore frequency to 51 Hz, use governer set point to $51 + (51 - 50.43) = 51.57\ Hz$

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